

Programming L^AT_EX —

A survey of documentation and packages

Brian Dunn

bd@BDTechConcepts.com

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Abstract

A survey of documentation for L^AT_EX. Included are references to printed and electronic books and manuals, symbol lists, FAQs, the L^AT_EX source code, CTAN and distributions, programming-related packages, users groups and online communities, and information on creating packages and documentation.

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Introduction

Reinventing the wheel may be useful if you think that you can do it better. Worse, though, is not even being aware that the wheel has already been invented in the first place, which can be an embarrassing waste of time. Such can be the case both for a new L^AT_EX programmer who isn't aware of the many ways things may be done, but also for someone, this author included, who learned L^AT_EX many years ago but may have missed some of the recent advancements in package code and documentation.

A wealth of information is available, not only in print and online, but also directly embedded in the typical L^AT_EX distribution. The following is meant to be a broad overview of some of today's resources for L^AT_EX programmers.

In some cases the same document may be listed in several categories. For example, a graphics FAQ also available in French may be listed under graphics, FAQs, and also French documents.

Many older documents are not included.

(The latest version of this document is available as the L^AT_EX docsurvey package.)

Printed books

Even in an electronic/online era, printed books still have the advantage of being able to be opened for reference without taking up space on the screen. Printed books also provide extended discussion of useful topics, have extensive human-edited indexes which are more useful than a simple document-wide search function, and some are also available in electronic format.

T_EX FAQ

T_EX FAQ. URL: <https://texfaq.org/>.

An online resource, which includes a detailed list of printed books.

More Math Into L^AT_EX

Grätzer

George Grätzer. *More Math Into L^AT_EX*. 5th ed. Springer, 2016. ISBN: 978-3-319-23795-4. URL: <https://www.springer.com/gp/book/9783319237954>.

Updated edition.

Guide to L^AT_EX

Kopka et al.

Helmut Kopka and Patrick W. Daly. *Guide to L^AT_EX*. 4th ed. Addison-Wesley Professional, 2004. 597 pp. ISBN: 0-321-17385-6. URL: <https://www.pearson.com/us/higher-education/program/Kopka-Guide-to-La-TeX-4th-Edition/PGM156755.html>.

An introduction and more advanced material, including an extensive reference guide.

L^AT_EX Beginner's Guide

Kottwitz

Stefan Kottwitz. *L^AT_EX Beginner's Guide*. Packt Publishing, 2011. ISBN: 1847199860. URL: <https://www.packtpub.com/product/latex-beginner-s-guide/9781847199867>.

An overview with numerous examples.

L^AT_EX Cookbook

Kottwitz

Stefan Kottwitz. *L^AT_EX Cookbook*. Packt Publishing, 2015. ISBN: 978-1-784-39514-8. URL: <http://latex-cookbook.net>.

More examples.

L^AT_EX: A Document Preparation System

Lamport

Leslie Lamport. *L^AT_EX: A Document Preparation System*. 2nd ed. Addison Wesley Professional, 1994. 272 pp. ISBN: 0-201-52983-1. URL: <https://www.pearson.com/us/higher-education/program/Lamport-La-TeX-A-Document-Preparation-System-2nd-Edition/PGM159713.html>.

The classic introduction to L^AT_EX, in continuous reprint for decades.

The L^AT_EX Companion

Mittelbach et al.

Frank Mittelbach et al. *The L^AT_EX Companion*. 2nd ed. Addison-Wesley, 2004. 1090 pp. ISBN: 0-201-36299-6. URL: <http://www.informit.com/store/latex-companion-9780133387667>.

Provides extended discussion and examples of the inner workings of L^AT_EX and numerous useful packages.

Books about T_EX, typography, and friends**T_EX Users Group**

T_EX Users Group. *Books about T_EX, typography, and friends*. URL: <http://tug.org/books/>.

T_EX Users Group book store, with reviews. Includes more than 75 books. Categories: published by TUG, by Donald E. Knuth, about T_EX and its applications, about typography and fonts, and about other related topics. Discounts for TUG members.

Presentations with L^AT_EX**VoSS**

Herbert VoSS. *Presentations with L^AT_EX*. Lehmanns Media GmbH, 2012. 206 pp. ISBN: 9783865414960.

PSTricks: Graphics and PostScript for T_EX and L^AT_EX**VoSS**

Herbert VoSS. *PSTricks: Graphics and PostScript for T_EX and L^AT_EX*. UIT Cambridge, 2011. ISBN: 978-1-906-86013-4. URL: <https://www.uit.co.uk/pstricks>.

Typesetting Mathematics with L^AT_EX**VoSS**

Herbert VoSS. *Typesetting Mathematics with L^AT_EX*. UIT Cambridge, 2010. ISBN: 978-1-906-86017-2. URL: <https://www.uit.co.uk/typesetting-mathematics-with-latex>.

Typesetting Tables with L^AT_EX**VoSS**

Herbert VoSS. *Typesetting Tables with L^AT_EX*. UIT Cambridge, 2011. ISBN: 978-1-906-86025-7. URL: <https://www.uit.co.uk/typesetting-tables-with-latex>.

Books and documentation by category

Most of these are provided with the T_EX distribution, and may be updated with each release. Access the embedded documentation from a command line using the `texdoc` program.

T_EX

For a list of older books, see <https://www.texfaq.org/FAQ-tex-books>.

T_EX for the Impatient**Abrahams et al.**

Paul W. Abrahams, Kathryn A. Hargreaves, and Karl Berry. *T_EX for the Impatient*. 2020. 393 pp. URL: <https://ctan.org/pkg/impatient>.

A tutorial and reference for T_EX, plain T_EX, and Eplain. Also available in French and Chinese.
(`texdoc impatient`).

A Gentle Introduction to T_EX**Doob**

Michael Doob. *A Gentle Introduction to T_EX. A Manual for Self-study*. 2002. 97 pp. URL: <https://ctan.org/pkg/gentle>.

A comprehensive tutorial on plain T_EX.
(`texdoc gentle`).

T_EX by Topic**Eijkhout**

Victor Eijkhout. *T_EX by Topic. A T_EXnician's Reference*. Addison-Wesley UK, 1991. 317 pp. ISBN: 0-201-56882-9. URL: <http://eijkhout.net/texbytopic/texbytopic.html>.

A reference for T_EX. This may be useful for understanding the source code of L^AT_EX packages, many of which are quite old and written in low-level T_EX. (texdoc texbytopic).

Wikibooks

Wikibooks. *T_EX*. URL: <https://en.wikibooks.org/wiki/Tex>.

An online book about low-level T_EX.

Getting Started with Plain T_EX**Wilkins**

D. R. Wilkins. *Getting Started with Plain T_EX*. 1994. 40 pp. URL: <http://www.ntg.nl/doc/wilkins/pllong.pdf>.

L^AT_EX**L^AT_EX 2_ε for authors****L^AT_EX 3 Project Team**

L^AT_EX 3 Project Team. *L^AT_EX 2_ε for authors*. 2020. 31 pp. URL: <https://ctan.org/pkg/usrguide>.

An overview of the new features of L^AT_EX 2_ε compared to L^AT_EX 2.09. (texdoc usrguide).

Writing Scientific Documents Using L^AT_EX**Bennieston**

Andrew J. Bennieston. *Writing Scientific Documents Using L^AT_EX*. 2009. URL: <https://ctan.org/pkg/intro-scientific>.

An introduction to typesetting scientific documents.

Formatting Information, A beginners introduction to typesetting with L^AT_EX**Flynn**

Peter Flynn. *Formatting Information, A beginners introduction to typesetting with L^AT_EX*. 2005. URL: <https://ctan.org/pkg/beginlatex>.

A beginners introduction to typesetting with L^AT_EX.

The very short guide to typesetting with L^AT_EX**Flynn**

Peter Flynn. *The very short guide to typesetting with L^AT_EX*. 2016. URL: <https://ctan.org/pkg/latex-veryshortguide>.

A four-page introduction.

L^AT_EX 2_ε: An unofficial reference manual**Greenwade et al.**

George D. Greenwade et al. *L^AT_EX 2_ε: An unofficial reference manual*. English, French, Spanish. 246 pp. URL: <https://latexref.xyz>.

A thorough but concise reference manual for L^AT_EX 2_ε, available in several languages.
(texdoc -l latex2e-help).

Getting something out of L^AT_EX**Hefferon**

Jim Hefferon. *Getting something out of L^AT_EX*. 2009. URL: <https://ctan.org/pkg/first-latex-doc>.

Create your first document in L^AT_EX.

Guide to L^AT_EX**Kopka et al.**

Helmut Kopka and Patrick W. Daly. *Guide to L^AT_EX*. 4th ed. Addison-Wesley Professional, 2004. 597 pp. ISBN: 0-321-17385-6. URL: <https://www.pearson.com/us/higher-education/program/Kopka-Guide-to-La-TeX-4th-Edition/PGM156755.html>.

An introduction and more advanced material, including an extensive reference guide.

L^AT_EX Beginner's Guide**Kottwitz**

Stefan Kottwitz. *L^AT_EX Beginner's Guide*. Packt Publishing, 2011. ISBN: 1847199860. URL: <https://www.packtpub.com/product/latex-beginner-s-guide/9781847199867>.

An overview with numerous examples.

L^AT_EX Cookbook**Kottwitz**

Stefan Kottwitz. *L^AT_EX Cookbook*. Packt Publishing, 2015. ISBN: 978-1-784-39514-8. URL: <http://latex-cookbook.net>.

More examples.

L^AT_EX: A Document Preparation System**Lamport**

Leslie Lamport. *L^AT_EX: A Document Preparation System*. 2nd ed. Addison Wesley Professional, 1994. 272 pp. ISBN: 0-201-52983-1. URL: <https://www.pearson.com/us/higher-education/program/Lamport-La-TeX-A-Document-Preparation-System-2nd-Edition/PGM159713.html>.

The classic introduction to L^AT_EX, in continuous reprint for decades.

Getting Started with L^AT_EX 2_ε**Morris**

Michael P. Morris. *Getting Started with L^AT_EX 2_ε*. 2020. 33 pp. URL: <https://ctan.org/pkg/startlatex2e>.

A beginner's bare-bones overview. (texdoc startlatex2e).

The Not So Short Introduction to L^AT_EX 2_ε**Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

L^AT_EX for Complete Novices**Talbot**

Nicola L. C. Talbot. *L^AT_EX for Complete Novices*. Dickimaw Books, 2012. 279 pp. URL: <http://www.dickimaw-books.com>.

An extensive introduction for a non-technical person. (texdoc dickimaw-novices).

Using L^AT_EX to Write a PhD Thesis**Talbot**

Nicola L. C. Talbot. *Using L^AT_EX to Write a PhD Thesis*. Dickimaw Books, 2013. 146 pp. URL: <http://www.dickimaw-books.com>.

A followup to *L^AT_EX for Complete Novices*, including extensive discussion about bibliographies, indexes, and glossaries. (texdoc dickimaw-thesis).

Wikibooks

Wikibooks. *LaTeX*. URL: <https://en.wikibooks.org/wiki/LaTeX>.

An online book, includes information about creating L^AT_EX packages and classes.

Lua^AT_EX**Lua^AT_EX Reference Manual****Lua^AT_EX development team**

Lua^AT_EX development team. *Lua^AT_EX Reference Manual*. 2020. 318 pp. URL: <http://mirrors.ctan.org/systems/doc/luatex/luatex.pdf>.

The complete reference. (texdoc luatex).

A guide to Lua^AT_EX**Pégourié-Gonnard**

Manuel Pégourié-Gonnard. *A guide to Lua^AT_EX*. 2013. 14 pp. URL: <https://ctan.org/pkg/lualatex-doc>.

An overview, and references to related packages. (texdoc luatex-doc).

X_ƎL^AT_EX**font-change-xetex****Dhawan**

Amit Raj Dhawan. *font-change-xetex. Macros to use OpenType and TrueType fonts with X_ƎL^AT_EX*. 2016. 21 pp. URL: <https://ctan.org/pkg/font-change-xetex>.

For plain X_ƎL^AT_EX. (texdoc font-change-xetex).

The X_YT_EX Companion**Goossens et al.**

Michel Goossens et al. *The X_YT_EX Companion. T_EX meets OpenType and Unicode*. 2010. 112 pp. URL: <https://ctan.org/pkg/xetex>.

Introduction to OpenType and Unicode, using OpenType fonts, handling Unicode-encoded sources.

The X_YT_EX reference guide**Robertson et al.**

Will Robertson, Khaled Hosny, and Karl Berry. *The X_YT_EX reference guide*. 2019. 25 pp. URL: <https://ctan.org/pkg/xetex>.

A summary of additional features over T_EX. (texdoc xetex-reference).

L^AT_EX3 and expl3**The L^AT_EX3 Interfaces****L^AT_EX3 Project Team**

L^AT_EX3 Project Team. *The L^AT_EX3 Interfaces*. 2020. 310 pp. URL: <https://ctan.org/pkg/l3kernel>.

Reference documentation for the expl3 programming environment. (texdoc interface3).

The L^AT_EX3 kernel: style guide for code authors**L^AT_EX3 Project Team**

L^AT_EX3 Project Team. *The L^AT_EX3 kernel: style guide for code authors*. 2020. 5 pp. URL: <https://ctan.org/pkg/l3kernel>.

Style guide for authors using expl3. (texdoc l3styleguide).

The expl3 package and L^AT_EX3 programming**L^AT_EX3 Project Team**

L^AT_EX3 Project Team. *The expl3 package and L^AT_EX3 programming*. 2020. 16 pp. URL: <https://ctan.org/pkg/l3kernel>.

Introduction to expl3. (texdoc expl3).

Bibliography**Tame the BeaST****Markey**

Nicolas Markey. *Tame the BeaST. The B to X of BibT_EX*. 2009. 48 pp. URL: <https://ctan.org/pkg/tamethebeast/>.

About bibliographies and BibT_EX. (texdoc tamethebeast).

Biblatex Cheat Sheet**Rees**

Clea F. Rees. *Biblatex Cheat Sheet*. 2017. 2 pp. URL: <https://ctan.org/pkg/biblatex-cheatsheet>.

A tri-fold quick reference. (texdoc biblatex-cheatsheet).

Math

Users Guide for the amsmath Package

American Mathematical Society et al.

American Mathematical Society and L^AT_EX3 Project Team. *Users Guide for the amsmath Package*. 2020. 44 pp. URL: <https://ctan.org/pkg/amsmath>.

How to use amsmath. Also see [International languages](#) for the Italian, Japanese, and Vietnamese translations. (texdoc amsmath).

Short Math Guide for L^AT_EX

Downes et al.

Michael Downes and Barbara Beeton. *Short Math Guide for L^AT_EX*. 2017. 21 pp. URL: <https://ctan.org/pkg/short-math-guide>.

A summary of features in L^AT_EX and packages for writing math formulas. (texdoc short-math-guide).

More Math Into L^AT_EX

Grätzer

George Grätzer. *More Math Into L^AT_EX*. 5th ed. Springer, 2016. ISBN: 978-3-319-23795-4. URL: <https://www.springer.com/gp/book/9783319237954>.

Updated edition.

Farbige Mathematik

VoSS

Herbert VoSS. “Farbige Mathematik”. German. In: *TeXnische Komödie* (2004). URL: <https://ctan.org/pkg/voss-mathcol>.

Math in color. In German, but with easy-to-use examples. (texdoc voss-mathcol).

Typesetting Mathematics with L^AT_EX

VoSS

Herbert VoSS. *Typesetting Mathematics with L^AT_EX*. UIT Cambridge, 2010. ISBN: 978-1-906-86017-2. URL: <https://www.uit.co.uk/typesetting-mathematics-with-latex>.

Page headings

The fancyhdr and extramarks packages

Oostrum

Pieter van Oostrum. *The fancyhdr and extramarks packages*. 2021. 74 pp. URL: <https://ctan.org/pkg/fancyhdr>.

Documents the fancyhdr and extramarks packages. Also includes an overview of the L^AT_EX page mark system. (texdoc fancyhdr).

Tables

Also see the T_EX FAQ Floats section: <https://www.texfaq.org/#floats>

Publication-quality tables in L^AT_EX

Fear

Simon Fear. *Publication-quality tables in L^AT_EX*. 2016. 18 pp. URL: <https://ctan.org/pkg/booktabs>.

Documents the booktabs package, and also includes thoughts on the design of tabular layouts in general.
(texdoc booktabs).

Typesetting Tables with L^AT_EX

VoSS

Herbert VoSS. *Typesetting Tables with L^AT_EX*. UIT Cambridge, 2011. ISBN: 978-1-906-86025-7. URL: <https://www.uit.co.uk/typesetting-tables-with-latex>.

Graphics

Also see the T_EX FAQ Graphics section: <https://www.texfaq.org/#graphics>

Visual PSTricks

Casteleyn

Jean Pierre Casteleyn. *Visual PSTricks*. English, French. 2016. 261 pp. URL: <https://ctan.org/pkg/visualpstricks>.

A visual FAQ consisting of a small example for each effect. (texdoc -l visualpstricks).

Visual TikZ

Casteleyn

Jean Pierre Casteleyn. *Visual TikZ*. English, French. 2018. 221 pp. URL: <https://ctan.org/pkg/visualtikz>.

A visual FAQ consisting of a small example for each effect. (texdoc -l visualtikz).

Using Imported Graphics in L^AT_EX and pdfL^AT_EX

Reckdahl

Keith Reckdahl. *Using Imported Graphics in L^AT_EX and pdfL^AT_EX*. 2006. 124 pp. URL: <https://ctan.org/pkg/epslatex>.

The TikZ and PGF Packages

Tantau

Till Tantau. *The TikZ and PGF Packages*. 2020. 1321 pp. URL: <https://ctan.org/pkg/pgf>.

As well as documenting the packages, this manual also includes “General guidelines and principles concerning the creation of graphics for scientific presentations, papers, and books”.
(texdoc pgfmanual).

PSTricks: Graphics and PostScript for T_EX and L^AT_EX

VoSS

Herbert VoSS. *PSTricks: Graphics and PostScript for T_EX and L^AT_EX*. UIT Cambridge, 2011. ISBN: 978-1-906-86013-4. URL: <https://www.uit.co.uk/pstricks>.

Music

L^AT_EX for Musicians

Gonzato

Guido Gonzato. *L^AT_EX for Musicians*. 2019. 66 pp. URL: <https://ctan.org/pkg/latex4musicians>.

Packages and programs for music symbols, lyrics, chord sheets, sheet music, and guitar tablature.
(texdoc latex4musicians).

Presentations

Beamer by Example

Mertz et al.

Andrew Mertz and William Slough. “Beamer by Example”. In: *The PracT_EX Journal* 2005.4 (2005). URL: <http://tug.org/pracjourn/2005-4/mertz/mertz.pdf>.

Graduated examples of the beamer package.

Examples from the book Presentations with L^AT_EX

VoSS

Herbert VoSS. *Examples from the book Presentations with L^AT_EX*. German. 2009. URL: <https://ctan.org/pkg/presentations>.

Source for examples from the book.

Examples from the book Presentations with L^AT_EX

VoSS

Herbert VoSS. *Examples from the book Presentations with L^AT_EX*. 2012. URL: <https://ctan.org/pkg/presentations-en>.

Source for examples from the book.

Presentations with L^AT_EX

VoSS

Herbert VoSS. *Presentations with L^AT_EX*. Lehmanns Media GmbH, 2012. 206 pp. ISBN: 9783865414960.

Fonts

Also see the T_EX FAQ Fonts section: <https://www.texfaq.org/#fonts>

The L^AT_EX Font Catalogue

Jørgensen

Palle Jørgensen. *The L^AT_EX Font Catalogue*. URL: <https://www.tug.org/FontCatalogue/>.

A detailed list of fonts for L^AT_EX, each with samples and setup information.

L^AT_EX font encodings

Mittelbach et al.

Frank Mittelbach et al. *L^AT_EX font encodings*. 2016. 39 pp. URL: <https://ctan.org/pkg/encguide>.

About T1 encoding, OT1, etc.

(texdoc encguide).

Essential NFSS2, version 2

Rahtz

Sebastian Rahtz. “Essential NFSS2, version 2”. In: *TUGBoat* 14.2 (1993), pp. 132–137. URL: tug.org/TUGboat/Articles/tb14-2/tb39rahtz-nfss.pdf.

A user’s view of the New Font Selection Scheme, version 2.

Using TrueType fonts with T_EX (L^AT_EX) and pdfT_EX (pdfL^AT_EX)**Rakityansky**

Damir Rakityansky. *Using TrueType fonts with T_EX (L^AT_EX) and pdfT_EX (pdfL^AT_EX)*. URL: <http://www.radamir.com/tex/ttf-tex.htm>.

Font selection in L^AT_EX: The most frequently asked questions**Schmidt**

Walter Schmidt. “Font selection in L^AT_EX: The most frequently asked questions”. In: *The PracT_EX Journal* 2006.1 (2006). URL: tug.org/pracjourn/2006-1/schmidt/schmidt.pdf.

Covers basic commands, default fonts, available font families, fonts for certain parts of the document.

L^AT_EX 2_ε font selection**Team**

L^AT_EX3 Project Team. *L^AT_EX 2_ε font selection*. 2020. 35 pp. URL: <https://ctan.org/pkg/fntguide>.

Documentation of commands for selecting fonts, as well as those for defining the data-structures used by the selection commands. (texdoc fntguide).

Fonts and T_EX**T_EX User’s Group**

T_EX User’s Group. *Fonts and T_EX*. URL: <http://tug.org/fonts/>.

A collection of links related to T_EX and fonts.

Cyrillic languages support in L^AT_EX**Volovich et al.**

Vladimir Volovich, Werner Lemberg, and L^AT_EX3 Project Team. *Cyrillic languages support in L^AT_EX*. 1999. 7 pp. URL: <https://ctan.org/pkg/cyrguide>.

Installation, usage, encodings. (texdoc cyrguide).

FAQs, symbol references, cheat sheets**Visual PStricks****Casteleyn**

Jean Pierre Casteleyn. *Visual PStricks*. English, French. 2016. 261 pp. URL: <https://ctan.org/pkg/visualpstricks>.

A visual FAQ consisting of a small example for each effect. (texdoc -l visualpstricks).

Visual TikZ**Casteleyn**

Jean Pierre Casteleyn. *Visual TikZ*. English, French. 2018. 221 pp. URL: <https://ctan.org/pkg/visualtikz>.

A visual FAQ consisting of a small example for each effect. (texdoc -l visualtikz).

L^AT_EX 2_ε Cheat Sheet**Chang**

Winston Chang. *L^AT_EX 2_ε Cheat Sheet*. 2006. 2 pp. URL: <https://ctan.org/pkg/latexcheat>.

A quick-reference guide for L^AT_EX and BibT_EX. Also in Brazilian Portuguese, German, Japanese, and Spanish. (texdoc latexcheat).

Detexify

Detexify. URL: <http://detexify.kirelabs.org/classify.html>.

Draw a symbol, and the website tells you which macros might make that symbol.

T_EX FAQ

T_EX FAQ. URL: <https://texfaq.org/>.

An online resource, which includes a detailed list of printed books.

Online tutorials on L^AT_EX

Indian T_EX Users Group

Indian T_EX Users Group. *Online tutorials on L^AT_EX*. 2000. URL: <http://tug.org/tutorials/tugindia/>.

An extensive tutorial covering many aspects of L^AT_EX.

L^AT_EX Cheat Sheet

Lammarsch

Marion Lammarsch. *L^AT_EX Cheat Sheet*. 2017. 4 pp. URL: <https://ctan.org/pkg/latex-refsheet>.

A reference for L^AT_EX with KOMA-Script. (texdoc latex-refsheet).

The Comprehensive L^AT_EX Symbol List

Pakin

Scott Pakin. *The Comprehensive L^AT_EX Symbol List*. 2017. 348 pp. URL: <https://ctan.org/pkg/comprehensive>.

More than 14,000 symbols and L^AT_EX commands. (texdoc comprehensive).

The Visual L^AT_EX FAQ

Pakin

Scott Pakin. *The Visual L^AT_EX FAQ*. 33 pp. URL: <https://ctan.org/pkg/visualfaq>.

Click on a visual element to learn how it is programmed. (texdoc visualFAQ).

Biblatex Cheat Sheet

Rees

Clea F. Rees. *Biblatex Cheat Sheet*. 2017. 2 pp. URL: <https://ctan.org/pkg/biblatex-cheatsheet>.

A tri-fold quick reference. (texdoc biblatex-cheatsheet).

Every symbol (most symbols) defined by unicode-math

Robertson

Will Robertson. *Every symbol (most symbols) defined by unicode-math*. 2018. 119 pp. URL: <https://ctan.org/pkg/unicode-math>.

Unicode math symbols. (texdoc unimath-symbols).

T_EX font errors: Cheatsheet

Schlömer

Nico Schlömer. *T_EX font errors: Cheatsheet*. 2010. 3 pp. URL: <https://ctan.org/pkg/tex-font-errors-cheatsheet>.

How T_EX integrates fonts, and related error messages. (texdoc tex-font-errors-cheatsheet).

shapecatcher

shapecatcher. URL: <http://shapecatcher.com/>.

Draw a symbol, and the website tells you which Unicode symbols it might be.

T_EX Resources on the Web

T_EX Users Group

T_EX Users Group. *T_EX Resources on the Web*. URL: <http://tug.org/interest.html>.

A large collection of links to numerous resources.

Source code

The source code for L^AT_EX 2_ε itself is also included in the distribution.

The L^AT_EX 2_ε Sources

Braams et al.

Johannes Braams et al. *The L^AT_EX 2_ε Sources*. 955 pp. URL: <https://ctan.org/pkg/source2e>.

Occasionally useful for figuring out how something really works. (texdoc source2e).

List of internal L^AT_EX 2_ε Macros useful to Package Authors

Scharrer

Martin Scharrer. *List of internal L^AT_EX 2_ε Macros useful to Package Authors*. 14 pp. URL: <https://ctan.org/pkg/macros2e>.

A list of the core L^AT_EX macros, each of which is linked to the source code. (texdoc macros2e).

International languages

Multiple languages

The following are available in several languages. Also see CTAN's topic for each language for additional translations of package and other documentation.

Free Programming Books

Foundation

Ebook Foundation. *Free Programming Books*. URL: <https://github.com/EbookFoundation/free-programming-books>.

A variety of T_EX-related and other programming books and documents.

L^AT_EX 2_ε: An unofficial reference manual**Greenwade et al.**

George D. Greenwade et al. *L^AT_EX 2_ε: An unofficial reference manual*. English, French, Spanish. 246 pp. URL: <https://latexref.xyz>.

A thorough but concise reference manual for L^AT_EX 2_ε, available in several languages.
(texdoc -l latex2e-help).

The Not So Short Introduction to L^AT_EX 2_ε**Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

Brazilian Portuguese**Guia Rápido L^AT_EX 2_ε****Chang**

Winston Chang. *Guia Rápido L^AT_EX 2_ε*. Brazilian. Trans. by Silvio C. G. Granja. 2006. 2 pp. URL: <https://ctan.org/pkg/latexcheat-ptbr>.

A quick-reference guide for L^AT_EX and BibT_EX. (texdoc latexcheat-ptbr).

L^AT_EX 2_ε Via Exemplos**Massago**

Sadao Massago. *L^AT_EX 2_ε Via Exemplos*. Brazilian. 2018. 264 pp. URL: <https://ctan.org/pkg/latex-via-exemplos>.

A study course.

Chinese

(Also see the Chinese category of the package list: p. 25)

T_EX 急就帖**Abrahams et al.**

Paul W. Abrahams, Kathryn A. Hargreaves, and Karl Berry. *T_EX 急就帖*. Chinese. 2014. 429 pp. URL: <https://ctan.org/pkg/impatient>.

A tutorial and reference for T_EX, plain T_EX, and Eplain. (texdoc impatient-cn).

Asymptote 范例教程**刘海洋**

刘海洋. *Asymptote 范例教程*. Chinese. 2009. URL: <https://ctan.org/pkg/asymptote-by-example-zh-cn>.

A tutorial for asymptote in the form of a graphical FAQ. (texdoc asymptote-by-example-zh-cn).

CTEXFAQ (常见问题集)**吴凌云**

吴凌云. *CTEXFAQ (常见问题集)*. Chinese. 2007. URL: <https://ctan.org/pkg/ctex-faq>.

FAQ from the Chinese T_EX Society. (texdoc ctex-faq).

一份简短的关于 L^AT_EX 安装的介绍

王然

王然. 一份简短的关于 L^AT_EX 安装的介绍. Chinese. 2020. 49 pp. URL: <https://ctan.org/pkg/install-latex-guide-zh-cn>.

Installing L^AT_EX and compiling documents, using various operating systems.
(texdoc install-latex-guide-zh-cn).

Asymptote 中的常见问题 (FAQ)

译者

译者. *Asymptote 中的常见问题 (FAQ)*. Chinese. 2009. 26 pp. URL: <https://ctan.org/pkg/asymptote-faq-zh-cn>.

A translation of the Asymptote FAQ. (texdoc asymptote-faq-zh-cn).

L^AT_EX Notes v 1.20

Huang

Alpha Huang. *L^AT_EX Notes v 1.20*. Chinese. 2008. 107 pp. URL: <https://ctan.org/pkg/latex-notes-zh-cn>.

An introduction to T_EX and L^AT_EX, including the use of Chinese fonts. (texdoc latex-notes).

French

Also see [Online communities](#).

TEX pour l'Impatient

Abrahams et al.

Paul W. Abrahams, Kathryn A. Hargreaves, and Karl Berry. *TEX pour l'Impatient*. French. 2004. 407 pp. URL: <https://ctan.org/pkg/impatient>.

A tutorial and reference for T_EX, plain T_EX, and Eplain. (texdoc impatient-fr).

Apprends L^AT_EX!

Baudoin

Marc Baudoin. *Apprends L^AT_EX!* French. 2012. 222 pp. URL: http://www.babafou.eu.org/Apprends_LaTeX.

A full textbook written for École Nationale Supérieure de Techniques Avancées.

Initiation à L^AT_EX

Bouzigues

Adrien Bouzigues. *Initiation à L^AT_EX. Pour débutants ou jeunes utilisateurs*. French. 2017. 273 pp. URL: <https://ctan.org/pkg/guide-latex-fr>.

A guide on L^AT_EX for beginners or advanced users.

Visual PSTricks

Casteleyn

Jean Pierre Casteleyn. *Visual PSTricks*. English, French. 2016. 261 pp. URL: <https://ctan.org/pkg/visualpstricks>.

A visual FAQ consisting of a small example for each effect. (texdoc -l visualpstricks).

Visual TikZ
Casteleyn

Jean Pierre Casteleyn. *Visual TikZ*. English, French. 2018. 221 pp. URL: <https://ctan.org/pkg/visualtikz>.

A visual FAQ consisting of a small example for each effect. (texdoc -l visualtikz).

French FAQ of the Gutenberg T_EX user group

French FAQ of the Gutenberg T_EX user group. French. Trans. by Marie-Paule Kluth. URL: <https://ctan.org/pkg/faq-fr>.

Rédaction avec L^AT_EX
Goulet

Vincent Goulet. *Rédaction avec L^AT_EX*. French. 2020. 196 pp. URL: <https://ctan.org/pkg/formation-latex-ul>.

An introductory course prepared for Université Laval, Québec, Canada.
(texdoc formation-latex-ul), (texdoc formation-latex-ul-diafos).

L^AT_EX 2_ε: An unofficial reference manual
Greenwade et al.

George D. Greenwade et al. *L^AT_EX 2_ε: An unofficial reference manual*. English, French, Spanish. 246 pp. URL: <https://latexref.xyz>.

A thorough but concise reference manual for L^AT_EX 2_ε, available in several languages.
(texdoc -l latex2e-help).

Tout Ce Que Vous Avez Toujours Voulus Savoir Sur L^AT_EX Sans Jamais Oser Le Demander
Lozano

Vincent Lozano. *Tout Ce Que Vous Avez Toujours Voulus Savoir Sur L^AT_EX Sans Jamais Oser Le Demander*. French. 2011. 271 pp. URL: <http://lozzone.free.fr/index.php?vlunch=latex>.

A book for beginners.

The Not So Short Introduction to L^AT_EX 2_ε
Oetiker

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

Utilisation de Graphiques Importés dans L^AT_EX 2
Reckdahl

Keith Reckdahl. *Utilisation de Graphiques Importés dans L^AT_EX 2*. French. 2001. 148 pp. URL: <https://ctan.org/pkg/fepslatex>.

How to import graphics in L^AT_EX 2_ε. (texdoc fepslatex).

X₃L^AT_EX Appliqué Aux Sciences Humaines**Rouquette**

Maïeul Rouquette. *X₃L^AT_EX Appliqué Aux Sciences Humaines*. French. 2012. 268 pp. URL: <https://ctan.org/pkg/latex-sciences-humaines>.

Apprendre à programmer en T_EX**Tellechea**

Christian Tellechea. *Apprendre à programmer en T_EX*. French. 2014. 580 pp. URL: <https://ctan.org/pkg/apprendre-a-programmer-en-tex>.

Basic programming of T_EX, with examples. (texdoc `apprendre`).

TeXniques

TeXniques. *Groupe francophone des Utilisateurs de TEX, L^AT_EX et logiciels compagnons*. French. URL: <https://www.gutenberg.eu.org/TeXniques>.

A collection of resources.

German

Also see [Users groups](#), and [Online communities](#).

L^AT_EX 2_ε Befehlsübersicht**Chang**

Winston Chang. *L^AT_EX 2_ε Befehlsübersicht*. German. Trans. by Tammo Schwindt. 2006. 2 pp. URL: <https://ctan.org/pkg/latexcheat-de>.

A quick-reference guide for L^AT_EX and BibT_EX. (texdoc `latexcheat-de`).

The Not So Short Introduction to L^AT_EX 2_ε**Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc `-l lshort`).

The DANTE T_EX Users Group Frequently Asked Questions

The DANTE T_EX Users Group Frequently Asked Questions. German. URL: <https://ctan.org/pkg/faq-de>.

Farbige Mathematik**VoSS**

Herbert VoSS. “Farbige Mathematik”. German. In: *TeXnische Komödie* (2004). URL: <https://ctan.org/pkg/voss-mathcol>.

Math in color. In German, but with easy-to-use examples. (texdoc `voss-mathcol`).

Anleitung**Weissenburger et al.**

Jens Weissenburger and Damir Rakityansky. *Anleitung*. German. 2003. URL: <https://ctan.org/pkg/anleitung>.

Using L^AT_EX, MikT_EX, and TrueType fonts.

Indian**A practical guide to L^AT_EX and polyglossia for Indian Languages****Holkar**

Rohit Dilip Holkar. *A practical guide to L^AT_EX and polyglossia for Indian Languages*. Marathi. 2017. 37 pp. URL: <https://ctan.org/pkg/latex-mr>.

Discusses Marathi, but also relevant to other Indian languages. (texdoc latex-mr).

Italian**Manuale utente per il pacchetto amsmath****American Mathematical Society et al.**

American Mathematical Society and L^AT_EX3 Project Team. *Manuale utente per il pacchetto amsmath*. Italian. Trans. by Giulio Agostini et al. 1999. 39 pp. URL: <http://tug.ctan.org/tex-archive/info/italian/amsl/doc/itamsl.doc.pdf>.

Manual for amsmath. (texdoc amsl/doc-it).

The Not So Short Introduction to L^AT_EX 2_ε**Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

Japanese

(Also see the Japanese category of the package list: p. 26)

L^AT_EX 2_ε for authors**L^AT_EX3 Project Team**

L^AT_EX3 Project Team. *L^AT_EX 2_ε for authors*. Japanese. Trans. by Yukitoshi FUJIMURA. 2015. 34 pp. URL: https://www.latex-project.org/help/documentation/usrguide_jpn.pdf.

An overview of the new features of L^AT_EX 2_ε compared to L^AT_EX 2.09.

Users Guide for the amsmath Package**American Mathematical Society et al.**

American Mathematical Society and L^AT_EX3 Project Team. *Users Guide for the amsmath Package*. Japanese. Trans. by Yukitoshi FUJIMURA. 2018. 45 pp. URL: https://www.latex-project.org/help/documentation/amsl/doc_jpn.pdf.

Manual for amsmath.

pL^AT_EX 2_ε チートシート**Chang**

Winston Chang. *pL^AT_EX 2_ε チートシート*. Japanese. Trans. by Takuto Asakura. 2006. 2 pp. URL: <https://ctan.org/pkg/platexcheat>.

A quick-reference guide for L^AT_EX and BibT_EX. (texdoc platexcheat).

Short Math Guide for L^AT_EX**Downes et al.**

Michael Downes and Barbara Beeton. *Short Math Guide for L^AT_EX*. Japanese. Trans. by Yukitoshi FUJIMURA. 2017. 21 pp. URL: https://www.latex-project.org/help/documentation/short-math-guide_jpn.pdf.

The Not So Short Introduction to L^AT_EX 2_ε**Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

Korean

(See the Korean category of the package list: p. 26)

Portuguese**Introdução ao Uso do Preparador de Documentos L^AT_EX****Campani**

Carlos A. P. Campani. *Introdução ao Uso do Preparador de Documentos L^AT_EX*. Portuguese. 2011. 188 pp. URL: <https://ctan.org/pkg/cursolatex>.

A tutorial as PDF slides. (texdoc cursolatex).

The Not So Short Introduction to L^AT_EX 2_ε**Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

Spanish**CervanTeX (Spanish TeX Group) FAQ**

CervanTeX (Spanish TeX Group) FAQ. Spanish. URL: <https://ctan.org/pkg/faq-es>.

(texdoc es-tex-faq).

Acordeón para L^AT_EX 2_ε**Chang**

Winston Chang. *Acordeón para L^AT_EX 2_ε*. Spanish. Trans. by J. Luis Rivera. 2006. 2 pp. URL: <https://ctan.org/pkg/latexcheat-esmx>.

A quick-reference guide for L^AT_EX and BibT_EX. (texdoc latexcheat-esmx).

L^AT_EX 2_ε: An unofficial reference manual **Greenwade et al.**

George D. Greenwade et al. *L^AT_EX 2_ε: An unofficial reference manual*. English, French, Spanish. 246 pp. URL: <https://latexref.xyz>.

A thorough but concise reference manual for L^AT_EX 2_ε, available in several languages. (texdoc -l latex2e-help).

The Not So Short Introduction to L^AT_EX 2_ε **Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

Vietnamese

Hướng dẫn sử dụng gói amsmath **American Mathematical Society et al.**

American Mathematical Society and L^AT_EX 3 Project Team. *Hướng dẫn sử dụng gói amsmath*. Vietnamese. Trans. by Ky Anh. 1999. 36 pp. URL: <https://ctan.org/pkg/amslatexdoc-vietnamese>.

Manual for amsmath. (texdoc amsldoc-vn).

The Not So Short Introduction to L^AT_EX 2_ε **Oetiker**

Tobias Oetiker. *The Not So Short Introduction to L^AT_EX 2_ε*. Numerous languages. 2015. 153 pp. URL: <https://ctan.org/pkg/lshort>.

Covers introductory material, customizations, and a simple package. (texdoc -l lshort).

Journals

The PracT_EX Journal **T_EX Users Group**

T_EX Users Group. *The PracT_EX Journal*. URL: <http://tug.org/pracjourn/>.

The online journal of the T_EX Users Group. Twenty issues, from 2005–2012.

TUGBoat **T_EX Users Group**

T_EX Users Group. *TUGBoat*. URL: <http://tug.org/TUGboat/>.

The Communications of the T_EX Users Group. Published since 1980. Articles covering every aspect of T_EX.

Interviews

TUG Interview Corner **T_EX Users Group**

T_EX Users Group. *TUG Interview Corner*. URL: <http://tug.org/interviews/>.

A large collection of interviews and articles about people related to T_EX. Includes links to more than 250 lectures and other recordings by Donald Knuth, and various historical information.

Typesetting examples

- A large collection of examples: <https://texample.net/>
- A collection of small examples: <http://tug.org/texshowcase/>
- Excerpts from many books: <https://ctan.org/topic/book-ex>
- Entire books: <http://www.tsengbooks.com/>
- Discussion: <https://tex.stackexchange.com/questions/1319/showcase-of-beautiful-typography-done-in-tex-friends>
- Discussion: <https://tex.stackexchange.com/questions/281415/showcase-of-beautiful-invitations-in-tex>

General typesetting theory

Discussion about general typesetting theory, presented by various T_EX-related authors.

For a list of non-L^AT_EX-specific books, see <https://www.texfaq.org/FAQ-type-books>.

Package canoniclayout

Beccari

Claudio Beccari. *Package canoniclayout*. 2011. 8 pp. URL: <https://ctan.org/pkg/canoniclayout>.

Documentation for the canoniclayout package. Also includes ideas regarding text-block proportions. (texdoc canoniclayout).

Publication-quality tables in L^AT_EX

Fear

Simon Fear. *Publication-quality tables in L^AT_EX*. 2016. 18 pp. URL: <https://ctan.org/pkg/booktabs>.

Documents the booktabs package, and also includes thoughts on the design of tabular layouts in general. (texdoc booktabs).

KOMA-Script The Guide

Kohm

Markus Kohm. *KOMA-Script The Guide*. 2018. 565 pp. URL: <https://ctan.org/pkg/koma-script>.

Documentation for the KOMA-Script package. Also includes discussion about the page layout of a book. (texdoc typearea).

The Octavo Package

Revets

Stefan A. Revets. “The Octavo Package”. In: *TUGboat* 23 3/4 (2002), p. 269. URL: <https://ctan.org/pkg/octavo>.

Design principles and guidelines emulating books from the Renaissance. (texdoc octavo).

The TikZ and PGF Packages

Tantau

Till Tantau. *The TikZ and PGF Packages*. 2020. 1321 pp. URL: <https://ctan.org/pkg/pgf>.

As well as documenting the packages, this manual also includes “General guidelines and principles concerning the creation of graphics for scientific presentations, papers, and books”.

(texdoc pgfmanual).

A TUFTE-STYLE BOOK

The Tufte-LaTeX Developers

The Tufte-LaTeX Developers. *A TUFTE-STYLE BOOK*. 2015. 42 pp. URL: <https://ctan.org/pkg/tufte-latex>.

Documentation for the Tufte-L^AT_EX document classes. Also includes layout ideas from the books of Edward R. Tufte.

(texdoc tufte-latex).

A Few Notes on Book Design

Wilson

Peter Wilson. *A Few Notes on Book Design*. 1st ed. The Herries Press, Aug. 2009. 139 pp. URL: <https://ctan.org/pkg/memdesign>.

More than 100 pages of discussion about book design and typography.

(texdoc memdesign).

Accessing embedded information

texdoc and mthelp

A large amount of documentation is included in a T_EX distribution. For TeXLive distributions, package documentation can be accessed with the texdoc program. Enter “texdoc -l <name>” to search for matching package, file, or program names. In some cases the same document is available in both letter or A4 paper sizes, or in several languages. texdoc is also available online, with popular packages sorted by category.

(<http://www.texdoc.net/>)

For MikT_EX, the mthelp program accesses package documentation. Enter “mthelp <name>”.

kpsewhich

The program kpsewhich may be used to find out where a file is located. kpsewhich filename searches for and returns the path to the given filename.

kpsewhich can also return directories, such as:

```
kpsewhich -var-value TEXMFROOT
kpsewhich -var-value TEXMFDIST
kpsewhich -var-value TEXMFLOCAL
```

Some package authors choose not to include the source code in the package documentation. To view the source code:

1. To locate and read a package’s .sty file:

```
kpsewhich package.sty
```

Usually these files have their comments removed, so it is better to use the .dtx file instead.

2. The .dtx file is usually available, and will have the package’s source code.

```
kpsewhich package.dtx
```

If it is not installed on your local system, it will be necessary to download the .dtx file from CTAN (see the next section).

The comments are not yet typeset and so will not be as easily read.

3. To typeset the documentation with the source code, copy the .dtx file and any associated image files somewhere local and then look for \OnlyDescription in the source. This command tells the ltxdoc package not to print the source code.

4. Remove \OnlyDescription, then process the .dtx file with

```
pdflatex package.dtx
```

Barring unusual circumstances, this will create a new documentation .pdf file with the package source code included.

Obtaining packages — Comprehensive T_EX Archive Network (CTAN)

T_EXLive installations use the tlmgr program to obtain packages. MikT_EX installations automatically install packages as needed. Where T_EX is installed by an operating-system package manager, that manager should be used to install additional packages.

For custom installations, it may be necessary to manually install packages downloaded from the Comprehensive T_EX Archive Network (CTAN), which provides a master collection of packages. A search function is available, which is useful when you know the name of a package or its author, and a list of topics is also provided. There are so many topics, however, that finding the right topic can be a problem in itself. One useful method to find what you are looking for is to search for a related package you may already know about, then look at its description on CTAN to see what topics are shown for it. Selecting these topics then shows you related packages.

(<https://ctan.org/>)

Useful classes, packages, and programs

Use texdoc or mthelp to access information about each of the following.

General-use packages and classes

Classes:

memoir, koma-script

Page layout and headings:

fancyhdr, geometry, microtype, nowidow, titleps

Fonts:

font-change-xetex, fontspec, mathspec, unicode-math

Sectioning:

epigraph, fncychap, quotchap, sectionbreak, sectsty, titlesec, tocvsec2

Table of contents:

etoc, minitoc, multitoc, shorttoc, titletoc, tocbibind, tocdata, tocloft, tocvsec2

Title page:

authblk, titling

Front and back matter:

abstract, appendix

Indexing:

makeindex, xindy, xindex, gindex, hvindex, idxlayout, imakeidx, index, makeidx, splitidx, varindex, xindex

Glossary:

glossaries, nomencl

Bibliography:

bibtex, biblatex, custom-bib

Cross-referencing:

cleveref, hyperref, url, xr-hyper, xurl, zref

Foot notes, margin notes, page notes:

bigfoot, endheads, endnotes, footmisc, manyfoot, marginfit, marginfix, marginnote, pagenote, parnotes, sidenotes

Math:

amsmath, amssymb, breqn, mathtools, resizgather, nicematrix, scalerel, stackrel

Theorems:

amsthm, apxproof, ntheorem, shadethm,
theorem, thmbox, thmtools

Units and fractions:

nicefrac, siunitx, xfrac

Floats:

caption, dblfloatfix, endfloat,
fewerfloatpages, float, floatrow, hypcap,
keyfloat, newfloat, placeins, rotfloat, stfloats,
subcaption, subfig, subfloat, wrapfig

Tabular:

array, booktabs, colortbl, longtable, ltable,
multirow, supertabular, tabularx, tabulary,
threeparttable, threeparttablex, widetable,
xltabular, xtab

Graphics:

asymptote, curves, fitbox, graphicx, pict2e,
pstricks, tikz, xy

Color:

normalcolor, xcolor

Lists:

enumerate, enumitem, paralist

Minipages:

eqparbox, minibox, pbox, shapepar

Quotations and verse:

csquotes, epigraph, quoting, verse

Verbatim:

fancyvrb, fvextra, moreverb, shortvrb,
upquote, verbatim

Frames:

boxedminipage2e, fancybox, fbox, framed,
mdframed, niceframe, shadow, tcolorbox

Embellishments:

fancypar, fancytabs, fourier-orns, lettrine,
pgfornament, pst-vectorian, sectionbreak

Multi-column:

adjmulticol, multicol, multicolrule, vwcol

Margins:

fullwidth, hanging, midpage

Line numbering:

lineno

Algorithms and listings:

algorithm2e, algorithmicx, listings,
listingsutf8, minted

Acronyms:

acro, acronym

Ordinals:

engord, fmtcount, nth

Direct formatting:

cancel, ellipsis, embrac, enparen, hyphenat,
lips, lua-check-hyphen, luacolor, pdfcol,
pdfcolmk, pdfrender, realscripts, relsize,
seqsplit, soul, soulpos, soulutf8, stackengine,
textfit, thinsp, trimclip, truncate, ulem,
umoline, underscore, uspace, xellipsis

Symbols:

academicons, amssymb, bbding, chemgreek,
dingbat, euro, eurosym, fontawesome,
fontawesome5, fourier-orns, gensymb,
latexsym, marvosym, metalogo, metalogox,
pifont, textalpha, textcomp, textgreek,
typicons, xunicode

Files:

attachfile, attachfile2, hyperxmp, intopdf,
pdfpages, pdfx, xmpincl

Admonitions:

awesomebox, notes

Editorial:

changebar, changelog, changes, easy-todo,
easyReview, ed, errata, fixme,
fixmetodonotes, pdfcomment, pdfmarginpar,
todo, todonotes, tram, xexchangebar

Accessibility:

accessibility, accsupp, axessibility,
pdfcomment, repltext, tagpdf

Presentations:

beamer, powerdot

Multi-language:

babel, beamer-rl, bidi, polyglossia

Chinese / Japanese / Korean (CJK):

cjkpunct, xeCJK

Chinese:

ctex, upzhkinsoku, xpinyin, zhlineskip,
zhspacing

Japanese:

bxjscls, luatexja, platex, plautopatch,
tascmac, uplatex, zxjatype

Debug:

chkfloat, cmdtrack, dprogress, inputtrc,
lua-visual-debug, refcheck

Korean:

kotex, luatexko, xetexko.

Automatic compiling

The programs arara and latexmk automatically recompile as necessary to resolve all dependencies.

Converting to HTML and other document formats**Using T_EX to generate the HTML:**

The lwarp package and the tex4ht program each use native L^AT_EX to interpret the document and generate HTML. More of L^AT_EX is supported compared to the translators listed below.

lwarp package:

Supports hundreds of packages. Generates HTML, and provides indirect assistance for EPUB conversion and copy/paste into a word-processor. <https://ctan.org/pkg/lwarp>

tex4ht program:

Generates HTML, EPUB, ODT, and Docbook. <http://tug.org/tex4ht/>

Translators:

These systems use external programs to translate a subset of L^AT_EX syntax into HTML. Search for each on CTAN (<http://ctan.org>).

H^Ev^Ea:

<http://hevea.inria.fr/>

T_TH:

<http://hutchinson.belmont.ma.us/tth/>

GELLMU:

<http://www.albany.edu/~hammond/gellmu/>

L^AT_EX_{ML}:

<http://dlmf.nist.gov/LaTeXML/>

PlasTeX:

<https://github.com/tiarno/plastex>

L^AT_EX₂HTML:

<http://www.latex2html.org/> and <http://ctan.org/pkg/latex2html>

T_EX₂page:

<http://ds26gte.github.io/tex2page/index.html>

L^AT_EX math to HTML:

GladT_EX takes a L^AT_EX math expression and generates the corresponding HTML.

GladT_EX:

<http://humenda.github.io/GladTeX/>

Programming L^AT_EX

A number of packages are especially useful for L^AT_EX programmers:

(texdoc <packagename>)

xifthen: Conditionals.	ifplatform: Detect operating system.
etoolbox: A wide range of programming tools, often avoiding the need to resort to low-level T _E X.	xstring: String manipulation.
etextools: Adds to etoolbox. Strings, lists, and more.	keyval, xkeyval, kvsetkeys: Key/value arguments.
xparse: Define macros and environments with flexible argument types.	pgfkeys, pgfkeyx: Another form of key/value arguments.
environ: Process environment contents.	kvoptions: Key/value package options.
arrayjobx, fifo-stack, forarray, forloop, xfor: Programming arrays, stacks, and loops.	expl3: L ^A T _E X3 programming.
iftex: Detect T _E X engine.	l3keys, l3keys2e: Key/value for L ^A T _E X3.
	chktex: Locates typographic errors.
	CTAN topic macro-supp: An entire topic of useful programming macros.

Creating and documenting new packages

Documentation for those interested in creating their own package or class:

L^AT_EX 2_ε for class and package writers

L^AT_EX3 Project Team

L^AT_EX3 Project Team. *L^AT_EX 2_ε for class and package writers*. 33 pp. URL: <https://ctan.org/pkg/clsguide>.

Programming a package or class. (texdoc clsguide).

Rolling your own Document Class: Using L^AT_EX to keep away from the Dark Side

Flynn

Peter Flynn. “Rolling your own Document Class: Using L^AT_EX to keep away from the Dark Side”. In: *TUGboat* 28:1 (2007), pp. 110–123. URL: <http://tug.org/TUGboat/tb28-1/tb88flynn.pdf>.

An overview of the article class.

How to develop your own document class — our experience

Mansfield

Niall Mansfield. “How to develop your own document class — our experience”. In: *TUGboat* 29:3 (2008), pp. 356–361. URL: <http://tug.org/TUGboat/tb29-3/tb93mansfield.pdf>.

A comparison of developing class vs. package files.

The doc and shortvrb packages

Mittelbach

Frank Mittelbach. *The doc and shortvrb packages*. 64 pp. URL: <https://ctan.org/pkg/doc>.

Packages for documenting packages. (texdoc doc).

The DocStrip program

Mittelbach et al.

Frank Mittelbach et al. *The DocStrip program*. 61 pp. URL: <https://ctan.org/pkg/docstrip>.

The program which processes `.dtx` and `.ins` files to generate documentation and `.sty` files.
(`texdoc docstrip`).

Good things come in little packages: An introduction to writing `.ins` and `.dtx` files **Pakin**

Scott Pakin. “Good things come in little packages: An introduction to writing `.ins` and `.dtx` files”. In: *TUGboat* 29:2 (2008), pp. 305–314. URL: <http://tug.org/TUGboat/tb29-2/tb92pakin.pdf>.

How and why to create your own `.dtx` and `.ins` files.

How to Package Your L^AT_EX Package **Pakin**

Scott Pakin. *How to Package Your L^AT_EX Package*. 36 pp. URL: <https://ctan.org/pkg/dtxtut>.

A tutorial. (texdoc dtxtut).

Users groups

T_EX Users Group: <http://tug.org>

Lists of international users groups:

- <http://tug.org/usergroups.html>
- <https://ctan.org/lugs>
- <http://www.ntg.nl/lug/>

Online communities

English forums:

T_EX — L^AT_EX Stack Exchange: Almost any question has already been asked, and a quick web search will find answers, ranked by vote. <http://tex.stackexchange.com>

L^AT_EX Community: A traditional forum with quick replies to your questions
<http://www.latex-community.org>

German forums:

TeXwelt: <http://texwelt.de/wissen/>

goLaTeX: <http://golatex.de>

French forums:

TeXnique.fr: <http://texnique.fr>

Mailing lists: Several dozen, spanning a wide range of T_EX-related topics.
<http://tug.org/mailman/listinfo>

Newsgroup: `comp.text.tex`

Online editing and collaboration

Overleaf: Collaborative editing of L^AT_EX documents online. <https://www.overleaf.com/>

Distributions — L^AT_EX for various operating systems

TeXLive: http://tug.org/texlive	Unix and Windows
MiKTeX: https://miktex.org	Windows and Mac
proTeXt: http://tug.org/protext/	Windows
MacTex: http://tug.org/mactex/	Mac

Change log

2017/03/06: Initial version.

2017/10/04: Added users groups, mailing lists, distributions, LuaT_EX, X_YT_EX, chktex. Organization and formatting improvements.

2017/10/14: More information about accessing embedded documentation.

2018/01/18: Added texdoc.net.

2018/01/21: Added latex-veryshortguide, first-latex-doc, beginlatex, intro-scientific, guide-latex-fr.

2018/03/24: Added interface3, dickimaw-novices, dickimaw-thesis.

2018/04/01: Added TeXnique.fr.

2018/06/28: Added sections for non-English documents and general typesetting theory. Updated host and name for T_EX FAQ. Added latex-via-exemplos and Ebook Foundation free programming books.

2018/10/18: Updated URL for *L^AT_EX 2_ε: An unofficial reference manual*.

2020/12/14: Improved bibliography. Added categories for math and music; startlatex2e; items written in French, German, Indian, Italian, Japanese, Portuguese, Vietnamese; Dante users group.

2021/01/02: Now uses biblatex. Added mthelp, many international documents, and categories for FAQs and cheat sheets, graphics, tables, and fonts. Added lists of packages by category. Added Overleaf.

2021/01/09: Added several resources from TUG including journals and interviews, and more for the list of packages.